

BUSINESS CASE	
Proposed Project	Ecosystem Discovery Application A browser-based educational web application designed for children in Grades 2–3 (approximately ages 7–8) that introduces Saskatchewan freshwater animals through interactive gameplay, including a timed hint-based discovery game, a sticker-book collection system, and supporting educational videos. The application emphasizes basic ecosystem understanding and conservation awareness using a 2D cartoon-style interface. All educational content is written at an approximate Grade 2–3 reading level to ensure clarity and accessibility for the intended audience.
Date Produced	27 January 2026
Background	Environmental education for children in Grades 2–3 is often delivered through textbooks or static digital content, which can limit engagement and make ecosystem concepts difficult to understand. Freshwater ecosystems are among the most familiar and locally relevant environments for young learners, yet there are few age-appropriate, interactive tools that allow children to actively explore and engage with this content. This project implements a browser-based educational application focused on Saskatchewan freshwater ecosystems. The system provides an interactive discovery experience where children learn by selecting hints, identifying animals in a visual environment, and collecting them in a sticker book. Additional reinforcement is provided through short educational videos. The application combines visual interaction, simple game mechanics, and structured hints to support curiosity-driven learning while remaining aligned with the cognitive level of children aged 7–8.
Business Need/ Opportunity	There is a need for engaging age-appropriate educational tools that help children in Grades 2–3 actively learn about freshwater ecosystems and conservation. Traditional approaches rely heavily on passive learning, which can reduce understanding and retention at this developmental stage. This project presents an opportunity to enhance early environmental education by providing an interactive, web-based application tailored to the reading and comprehension level of young learners. By focusing on a clearly defined freshwater ecosystem and a limited set of regional animals, the project maintains scope control while delivering meaningful

	educational value. .
Options	Option 1: Develop the Ecosystem Educational Discovery Application • Design and implement a browser-based educational application • Focus exclusively on a Saskatchewan freshwater ecosystem • Provide an interactive game experience (hint-based matching with timer and rewards) • Include a sticker-book collection system and educational video content • Follow an MVC architecture using vanilla JavaScript (ES6 modules) • Integrate a backend database (Supabase) for dynamic animal and hint data • Deliver a clearly defined Minimum Viable Product (MVP) with core gameplay features Option 2: Do Nothing • Continue relying on existing static educational resources • No interactive or discovery-based learning experience • No improvement in student engagement or learning outcomes
Cost-Benefit Analysis	
Option 1: Develop the Application Costs: • Time and effort invested by team members • Learning and coordination required for design and development • Potential technical challenges during implementation Benefits: • Increased engagement for young learners • Improved understanding of freshwater ecosystems • Supports interactive and exploratory learning • Reinforces conservation awareness at an early age • Meets course learning objectives related to software engineering, MVC architecture, and project management • Produces a portfolio-quality academic project Option 2: Do Nothing Costs: • No development or implementation effort required Benefits: • Continued reliance on static learning materials may limit student engagement, reduce	

retention of ecosystem concepts, and provide no opportunity for interactive discovery-based learning.

Recommendation

It is recommended to proceed with **Option 1: Development of the Ecosystem Discovery Application.**

This option provides meaningful educational value by introducing children in Grades 2–3 to freshwater animals through interactive learning at an appropriate reading level. The project remains focused on a clearly defined ecosystem scope, aligns with course requirements, and offers both educational benefits for users and practical software engineering experience for the project team.